Project: Predicting Diabetes Disease Progression

Submission deadline: October 24, 2026

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1 Project Objective

The objective of this project is to build and evaluate machine learning models for predicting a quantitative measure of diabetes disease progression one year after the baseline measurement. The project uses the `load_diabetes` dataset available in the `scikit-learn` library.

Unlike a classification problem where the objective is to predict whether a patient belongs to a particular class, this problem is a **regression problem**. The target variable is continuous and represents the quantitative measure of disease progression. Therefore, the central modelling objective is to learn a function

$$f : \mathbb{R}^{10} \to \mathbb{R}$$

such that

$$\hat{y} = f(\mathbf{x})$$

provides an accurate estimate of the disease-progression measure y from the patient's baseline characteristics $\mathbf{x}$.

2 Dataset Description

The dataset contains observations for 442 individuals and 10 input features. The features describe demographic, physiological, and biochemical characteristics measured at baseline.

Feature	Description
`age`	Age of the individual
`sex`	Sex of the individual
`bmi`	Body Mass Index
`bp`	Average blood pressure
`s1`	Total serum cholesterol
`s2`	Low-density lipoproteins (LDL)
`s3`	High-density lipoproteins (HDL)
`s4`	Total cholesterol / HDL ratio
`s5`	Log-transformed serum triglycerides level
`s6`	Blood sugar level

Table 1: Features in the diabetes dataset

The target variable is a quantitative measure of disease progression approximately one year after the baseline measurements.

Thus, each observation can be represented as

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{i,10}), \qquad y_i \in \mathbb{R},$$

where $\mathbf{x}_i$ contains the baseline characteristics of individual i and y_i represents the corresponding disease-progression measure.

3 Research Questions

Rather than treating the project as simply an exercise in fitting a regression model, the following questions will guide the analysis:

1. Which baseline characteristics have the strongest statistical association with disease progression?
2. How well can a linear model explain the variation in the target?
3. Does regularization improve generalization when predictors are correlated?
4. Do nonlinear models capture relationships that a linear model cannot?
5. How sensitive are the conclusions to the choice of model?
6. Which model provides the best balance between predictive accuracy and interpretability?

These questions allow the project to focus on both **prediction** and **understanding of the underlying modelling assumptions**.

4 Modelling Strategy

Several regression models will be considered to understand the effect of model complexity.

4.1 Linear Regression

The baseline model will be ordinary least squares regression

4.2 Ridge Regression

Because some predictors may be correlated, Ridge regression will also be considered.

4.3 Lasso Regression

An important conceptual advantage of Lasso is that the ℓ_1 penalty can drive some coefficients exactly to zero. Therefore, it can simultaneously perform regularization and feature selection.

4.4 Feature-Engineered Regression

Create additional features from the given variables using transformations such as powers and interaction terms:

$$x_i^2, \quad x_i^3, \quad x_i x_j.$$

Using this expanded feature set, fit and compare all three regression models: **OLS Linear Regression, Ridge Regression, and Lasso Regression**. Investigate how feature engineering and increasing polynomial degree affect model performance, overfitting, and the coefficients selected by the three regression methods.

5 Training and Testing Strategy

The dataset will be divided into training and testing sets. The training set is used to estimate model parameters, while the test set is kept separate until the final evaluation.

Let

$$D = D_{\text{train}} \cup D_{\text{test}}, \qquad D_{\text{train}} \cap D_{\text{test}} = \emptyset.$$

Model selection and hyperparameter tuning should be performed using only the training data, preferably through cross-validation. The test set should be used only for the final assessment of generalization performance.

6 Final Objective

The final objective is not merely to obtain the smallest prediction error. Instead, the project aims to investigate how the representation of the features and the choice of regularization affect regression performance.

The central question is:

> *How do feature engineering, model complexity, and regularization affect the ability of OLS, Ridge, and Lasso regression to predict diabetes disease progression?*

Your report should connect exploratory data analysis, feature engineering, statistical modelling, regularization, model evaluation, and interpretation into a coherent regression study.